

ECE 491 Project Guideline

Description

You will design a neural-network-based wildfire/no wildfire classification project. You can work in a group.

The group should consist of up to 2 students. You can use any programming language (Python, MATLAB, C++, etc.) and any framework (PyTorch, TensorFlow, Keras, Caffe, MATLAB neural network toolbox, etc.) you want, but I strongly recommend you to use PyTorch, as it is used by most researchers nowadays.

Please download the dataset “fire.zip”. It contains about 10k images. You will random split them by 9:1 for training and testing.

Report

You should write a report in an academic format, for example:

<https://www.ieee.org/conferences/publishing/templates.html>

Your report should include

1. Introduction
2. Method description. You may present your model’s detail.
3. Experimental results. What is your accuracy, precision, recall, f1-score, confusion matrix, AUC, ROC curve?
4. Conclusion
5. Contribution of each member If you are working in a group
6. Reference
7. Appendix: Code list

If you are working in a group, only one student needs to make a submission, but please make sure both of your names are in the report.